SYSTEM DOCUMENTATION & CODE REFERENCE V2.0

ArogyaNext Admin & Clinical OS

Complete file-by-file specification of all source modules, data models, state workflows, Firebase Firestore schemas, real-time sync mechanisms, and user interactions.

22+	7	5	Real-Time
SOURCE FILES	SPECIALIZED PANELS	ROLE PERSONAS	FIRESTORE SYNC

1. High-Level System Architecture & Cross-Panel Data Flow

ArogyaNext is built as a multi-tenant cloud hospital operating system using **React 19**, **Vite 8**, **Tailwind CSS 4**, and **Firebase Firestore / Auth**. Every clinical workflow connects without manual paper handoffs.



2. Firebase Firestore Database Collections & Schema Design

COLLECTION	DOCUMENT ID KEY	KEY FIELDS & TYPES	PANEL USAGE & PURPOSE
users	auth.uid	uid, email, fullName, role, hospitalId, createdAt	Auth & profile management. Controls tab visibility and multi-

COLLECTION	DOCUMENT ID KEY	KEY FIELDS & TYPES	PANEL USAGE & PURPOSE
			tenant scoping.
hospitals	`hospitalId` (e.g. HOSP_X92K)	`hospitalId, hospitalName, createdAt`	Workspace tenancy master table. Super Admin creates/manages hospital tenants.
doctors	`doctor.uid`	`name, hospitalId, specialty, isActive`	Active medical staff directory used by Receptionist to assign incoming patients.
appointments	Auto-generated UUID	`tokenNumber, patientName, patientAge, symptoms, doctorName, hospitalId, status, isEmergency, appointmentDate, timeSlot, timestamp, prescription, pharmacyStatus, scannedPrescriptionText`	Heart of the application. Tracks the end-to-end lifecycle of patient consultations and medicine fulfillment.
doctor_status	`doctorName`	`status ('Active'|'Away'|'In Consultation'), currentToken, lastUpdated`	Global availability indicator for doctors, allowing reception and patients to see doctor availability.
queue	`hospitalId`	`activeDoctor, activeToken, status, lastUpdated`	Broadcast queue state synced across TV lobby monitors and companion devices.

ℹ️ **Multi-Tenancy Security Rule:** Every single query across reception, doctor cabins, pharmacy, and analytics strictly filters by `hospitalId == userProfile.hospitalId`. Data never leaks across clinics or hospital branches.

3. Root Configuration & Application Bootstrap Files

package.json

CONFIG & MANIFEST

Path:	/package.json	**Type:**	Node.js ES Module Manifest
Scripts:	dev, build, lint, preview	**Frameworks:**	React 19.2, Vite 8.0, Tailwind CSS 4.2

WHAT IT CONTAINS:

Declares project dependencies, build toolchains, and runtime dependencies. Key packages include: `firebase (v12.11)` for Firestore and Authentication, `lucide-react (v0.577)` for clinical icons, `@tailwindcss/vite` and `tailwindcss (v4.2)` for utility CSS, and `eslint (v9.39)` for React hooks linting.

RUNTIME BEHAVIOR & PURPOSE:

- **Development:** Launches the Vite dev server with Hot Module Replacement (HMR).
- **Production:** Compiles React 19 JSX into minified ES chunks in the `/dist` directory.
- **Zero bloated dependencies:** No heavy external UI component kits; built with native DOM APIs, Canvas, SpeechSynthesis, and Tailwind for performance.

vite.config.js

BUILD BUNDLER CONFIG

Path:	/vite.config.js	**Bundler:**	Vite 8.0
Plugins:	@vitejs/plugin-react, @tailwindcss/vite	**Output:**	Single Page Application (SPA)

WHAT IT CONTAINS & BEHAVIOR:

Configures the Vite compiler. Directly integrates `tailwindcss()` as a first-class Vite plugin (Tailwind v4's new engine) avoiding legacy PostCSS latency. Enables Fast Refresh on React state changes during development.

vercel.json

DEPLOYMENT ROUTING

Path:	/vercel.json	**Target Platform:**	Vercel Edge / Static CDN
Rule:	URL Rewrites to /index.html	**Behavior:**	SPA Deep-Link Routing Support

WHAT IT CONTAINS & BEHAVIOR:

Defines a rewrite rule mapping all routes `/(.*)` to `/index.html`. Ensures that when hospital staff refresh pages or access direct sub-paths on cloud hosting (Vercel), the server returns the SPA root rather than a 404 error.

index.html

HTML ENTRY POINT

Path:	/index.html	**DOM Mounting:**	
Script Module:	/src/main.jsx	**Icon:**	/favicon.svg

WHAT IT CONTAINS & BEHAVIOR:

Provides the initial HTML shell. Contains responsive viewport meta tags (`width=device-width, initial-scale=1.0`), links to the medical cross SVG favicon, and triggers module execution of `/src/main.jsx`.

src/main.jsx

REACT ROOT INITIALIZER

Path:	/src/main.jsx	**Imports:**	StrictMode, createRoot, ./index.css, ./App.jsx
Root Node:	document.getElementById('root')	**Mode:**	React 19 StrictMode

WHAT IT CONTAINS & BEHAVIOR:

Instantiates the React virtual DOM tree with `createRoot()`. Wraps `<App />` in `<StrictMode>` to detect side-effects, deprecations, and race conditions in development. Imports global styling from `./index.css`.

src/firebase.js

BACKEND CLOUD SDK INSTANCE

Path:	/src/firebase.js	**Exports:**	db (Firestore), auth (FirebaseAuth)
Firebase Project:	arogyanext-ab375	**Services:**	Firestore Realtime DB, Firebase Auth

WHAT IT CONTAINS & BEHAVIOR:

Initializes the Firebase App using credentials (API Key, authDomain, projectId, storageBucket, messagingSenderId). Exports the singletons `db` (cloud database) and `auth` (authentication service), which all panels throughout the application import for database reading and writing.

src/index.css

GLOBAL STYLES & ANIMATIONS

Path:	/src/index.css	**Framework:**	Tailwind CSS 4 (@import "tailwindcss")
Keyframes:	marquee, neon-pulse, call-attention, sound-bar-1..4	**Special Utility:**	Custom Dark Scrollbar, Touch Optimization

WHAT IT CONTAINS:

- **CSS Reset & Touch:** Sets `scroll-behavior: smooth`, disables 300ms mobile tap delays with `-webkit-tap-highlight-color: transparent`, and clamps horizontal overflow.
- **Modern Dark Scrollbar:** Styled webkit scrollbars with slate-900 background track and rounded slate-600 thumb.
- **`animate-marquee`:** 28-second continuous ticker animation that pauses on user hover. Used for emergency lobby news.
- **`animate-neon-pulse`:** Cycling emerald glow effect with box-shadows applied to active consulting tokens.
- **`animate-call-attention`:** Subtle 1.04x scale heartbeat pulse for active calling banners.
- **`animate-sound-1..4`:** Sound wave equalizer bars simulating real-time audio playback during voice announcements.

src/App.css

LEGACY CSS UTILITIES

Path:	/src/App.css	**Purpose:**	Isometric hero graphics, button focus outlines

WHAT IT CONTAINS & BEHAVIOR:

Provides supplemental styling for 3D isometric transforms, perspective layers, button accessibility outlines, and documentation next-step containers.

src/App.jsx

ROOT ORCHESTRATOR & VIEW SWITCHER

Path:	/src/App.jsx	**State Hooks:**	user, userProfile, currentView, isAuthenticating, showAuthScreen, mobileMenuOpen
Listeners:	onAuthStateChanged(auth)	**Tabs Handled:**	receptionist, doctor, smartPad, pharmacy, hospitalAdmin, tv, admin

WHAT IT CONTAINS:

Acts as the brain of the frontend application. It orchestrates user authentication, role-based access control (RBAC), multi-device responsive navigation, and conditional rendering of the appropriate clinical panel.

DETAILED RUNTIME BEHAVIOR & FLOW:

- **Auth Initialization:** While verifying auth tokens, renders a dark animated splash screen with a bouncing caduceus (⚕).
- **Public vs Staff Gate:** If no user is logged in, renders `<LandingPage />`. If the user clicks "Staff Login", toggles `showAuthScreen = true` to display `<Auth />` with a back button.
- **Role Routing on Login:** Upon login, queries `users/{uid}` from Firestore. Dynamically defaults the active view based on `profile.role`: *Doctor* → Doctor Workspace, *Pharmacist* → E-Pharmacy, *Hospital Admin* → Analytics Hub, *Super Admin* → Founder Console, *Receptionist* → Front Desk.
- **Role-Based Navigation Tabs:** Filters the master tab array so staff can only see and access authorized modules. For example, doctors see Doctor & Smart Pad; pharmacists see Pharmacy & TV; Super Admins see all 7 modules.
- **Mobile UI Support:** Features dual mobile navigation: an interactive hamburger drawer and a horizontal scrollable pill bar, ensuring effortless one-thumb operation on smartphones.
- **Header Bar:** Displays the hospital access code (e.g. `HOSP_X92K`), authenticated staff name, role badge, and a one-click `signOut()` action.

src/Auth.jsx

STAFF AUTHENTICATION PORTAL

Path:	/src/Auth.jsx	**State Hooks:**	isLogin, isForgotPassword, email, password, fullName, hospitalId, role, showPassword, isLoading, error
Firebase APIs:	signInWithEmailAndPassword, createUserWithEmailAndPassword, sendPasswordResetEmail, setDoc		

WHAT IT CONTAINS & DETAILED BEHAVIOR:

- **Tri-Mode Form:** Smoothly switches between *Sign In*, *Create Staff Account*, and *Forgot Password*.
- **Hospital ID Validation:** When creating a staff account (unless role is Super Admin), checks if `hospitals/{hospitalId.toUpperCase()}` exists in Firestore. If invalid, blocks signup and alerts the user.
- **Automatic Doctor Directory Registration:** When a user with the role "Doctor" registers, the component creates both a `users/{uid}` document AND a `doctors/{uid}` record so front-desk staff can assign patients to them immediately.
- **Security & UX:** Features an eye icon to toggle password visibility, uppercase auto-formatting for hospital codes, and self-clearing error banners with friendly messages.

src/LandingPage.jsx

PUBLIC SHOWCASE & MARKETING

Path:	/src/LandingPage.jsx	**Sections:**	Hero, Features Grid (OCR, Voice, Triage), Workflow, Pricing, Demo Modal
Props:	onLoginClick (triggers staff login modal)		

WHAT IT CONTAINS & BEHAVIOR:

A landing page highlighting ArogyaNext's capabilities:

- **On-Device Prescription OCR:** Highlights Google ML Kit text recognition for converting old paper prescriptions to digital notes.
- **Multi-Language Voice AI:** Details regional speech triage supporting 6 Indian languages (Hindi, Tamil, Telugu, Marathi, Bengali, English).
- **Interactive Demo Lead Form:** Built-in modal allowing hospital administrators to request a custom clinic deployment.
- **Pricing & Commercial Calculator:** Showcases the SaaS pricing tier (₹999/month base + ₹500/doctor) with a 6-month free pilot promotion.

src/ReceptionistPanel.jsx

FRONT DESK CONTROL CENTER

Path:	/src/ReceptionistPanel.jsx	**Props:**	hospitalId
State Hooks:	patientName, patientAge, symptoms, selectedDoctor, isEmergency, appointmentDate, timeSlot, doctorsList, todaysPatients	**Firestore Listeners:**	onSnapshot(appointments where status == 'Waiting' & date == today)

DETAILED BEHAVIOR & LOGIC:

- **Doctor Sync:** On mount, queries `doctors` filtered by `hospitalId` to populate the doctor dropdown with real names and specialties.
- **Smart Token Generation:** Normal appointments generate `S-XX` tokens (e.g. S-42). Emergency walk-ins generate `E-XX` tokens (e.g. E-12) and bypass time slot requirements with status set to `IMMEDIATE`.
- **Live Grouped Queues:** Groups today's waiting appointments by doctor cabin in real-time. Receptionists can monitor workload distribution across all cabins simultaneously.
- **Appointment Cancellation:** Receptionists can cancel an appointment with a single tap, updating the Firestore status to `Cancelled` and removing the patient from doctor queues.
- **OCR Indicator:** If a patient booked via mobile with an uploaded prescription, an 📷 `OCR Rx` badge appears alongside their token.

src/DoctorPanel.jsx

DOCTOR CLINICAL WORKSPACE

Path:	/src/DoctorPanel.jsx	**Props:**	hospitalId, doctorName
State Hooks:	waitingPatients, currentPatient, prescriptionText, currentStatus ('Active'|'Away')	**Updates:**	appointments, doctor_status

DETAILED BEHAVIOR & LOGIC:

- **Doctor-Scoped Waiting Queue:** Real-time snapshot listener listening only for patients assigned to this doctor with status `Waiting`. Emergency patients flash red at the top.
- **Call In Action:** When the doctor clicks "Call In", the appointment status changes to `In Consultation` and the doctor's document in `doctor_status` updates with `currentToken`, triggering notifications across the TV display and companion apps.
- **Scanned Paper Rx (ML Kit OCR) Review:** Displays the patient's scanned previous prescription text in a monospaced preview card. The doctor can click "+ *Insert into E-Prescription*" to append those notes into the digital prescription.
- **Finish Consultation & E-Pharmacy Sync:** Doctor enters medicine notes and clicks "Finish Consultation". This sets appointment status to `Completed` and, if medicines were typed, sets `pharmacyStatus = 'Pending'`, routing the prescription to the hospital pharmacy counter.

src/SmartPrescriptionPanel.jsx

SMART LETTERHEAD & STYLUS SIGNATURE

Path:	/src/SmartPrescriptionPanel.jsx	**Props:**	hospitalId, doctorName
Key Tech:	HTML5 Canvas (2D Context), LocalStorage Base64, CSS Print Media	**Controls:**	Margin Top slider, Signature Bottom slider, Mask Lines toggle

DETAILED BEHAVIOR & LOGIC:

- **Custom Letterhead Upload:** Doctors can upload a PNG/JPEG image of their clinic letterhead. The file is converted into Base64 and stored in `localStorage` per doctor, loading instantly on subsequent visits.
- **Pixel-Perfect Calibrated Alignment:** Doctors use sliders to align where patient info and prescription text appear over their printed letterhead lines, and can toggle "Mask Background Lines" to prevent text collision.
- **Stylus Signature Pad:** Uses a 2x retina-scaled HTML5 Canvas with pointer event listeners (`onPointerDown`, `onPointerMove`, `onPointerUp`) supporting Apple Pencil, Surface Pen, or touch signatures.
- **Print & Pharmacy Sync:** When clicking "Finish, Send & Print", the component updates Firestore (routing to pharmacy) and invokes `window.print()`. Using print CSS, the UI toolbars disappear and only the letterhead prescription prints.

src/TVDisplay.jsx

LOBBY TV SIGNAGE & VOICE ANNOUNCEMENTS

Path:	/src/TVDisplay.jsx	**Props:**	hospitalId
APIs Used:	Web Speech API (speechSynthesis), Firestore Real-Time Query	**Visuals:**	Blinking Digital Clock, Neon Active Token Glow, Audio Equalizer Bars, Marquee Notice

DETAILED BEHAVIOR & LOGIC:

- **Full Screen TV Layout:** Designed for 55"+ smart TVs mounted in hospital waiting lobbies. Displays high-contrast text readable from across the room.
- **Multi-Doctor Cabin Grid:** Dynamically splits into side-by-side cards for each doctor on duty. Left half shows the active token in glowing emerald font; right half shows the next 4 patients in line.
- **Speech Synthesis (TTS) Engine:** When any doctor calls a token, an utterance is synthesized: "*Token number S 14, please proceed to Dr. Sharma's cabin.*" Accompanied by a floating announcement banner and animated equalizer bars.
- **Continuous Broadcast Marquee:** Bottom ticker displays clinic notices, reminding patients about token announcements and emergency procedures.

src/PharmacyPanel.jsx

E-PHARMACY FULFILLMENT COUNTER

Path:	/src/PharmacyPanel.jsx	**Props:**	hospitalId
Firestore Listener:	appointments where hospitalId == id & pharmacyStatus == 'Pending'		

DETAILED BEHAVIOR & LOGIC:

- **Real-Time Medicine Queue:** Real-time listener for any appointment in the facility where `pharmacyStatus == 'Pending'`. When a doctor finishes a consultation, the order appears immediately.
- **Prescription Cards:** Displays patient name, prescribing doctor, emergency badge (if urgent), and full medicine list.
- **Fulfillment Action:** Pharmacist prepares medicines and clicks "Mark as Dispensed ✅", updating the record to `pharmacyStatus = 'Dispensed'` and clearing the card from the active queue.

src/HospitalAdmin.jsx

DIRECTOR ANALYTICS & CSV EXPORT

Path:	/src/HospitalAdmin.jsx	**Props:**	hospitalId
Metrics:	Patients Today, Pharmacy Revenue, Avg Wait Time, Active Doctors	**Exports:**	CSV Reports (1 Day, 7 Days, 1 Month, 6 Months, 1 Year)

DETAILED BEHAVIOR & LOGIC:

- **Hospital Director KPI Matrix:** Displays four operational metric tiles with trend percentages and status indicators.
- **CSV Report Generator:** Queries historical appointments from Firestore for the selected timeframe. Formats timestamps, token IDs, doctor names, consultation status, and pharmacy fulfillment into CSV format, then triggers an automatic browser download.

src/AdminPanel.jsx

FOUNDER CONSOLE (SUPER ADMIN)

Path:	/src/AdminPanel.jsx	**Role:**	Restricted to Super Admin ('SYSTEM_MASTER')
Actions:	Create Hospital, Delete Hospital, Delete Staff Member		

DETAILED BEHAVIOR & LOGIC:

- **Platform-Wide Overview:** Summarizes total active hospitals, total registered doctors, and total staff members across the platform.
- **Hospital Workspace Generator:** Generates unique alphanumeric access codes (e.g. `HOSP_9X8B2`) and writes a document to `hospitals`.
- **Staff Account Governance:** Lists all registered staff members under each hospital tenant. Super Admin can remove compromised accounts with a single click.

src/TabletDoctorPanel.jsx

STYLUS FREEHAND TABLET PAD

Path:	/src/TabletDoctorPanel.jsx	**Purpose:**	Tablet-optimized freehand sketching with simulated AI OCR translation

WHAT IT CONTAINS & BEHAVIOR:

A dedicated tablet workspace featuring physical notepad ruling lines on HTML5 Canvas. Doctors write freehand prescriptions with a stylus; clicking "*Process Handwriting (OCR)*" translates the handwriting into editable digital text, which can be sent directly to the pharmacy.

4. Static Assets & Public Files

FILE PATH	TYPE / FORMAT	ROLE IN APPLICATION
`public/favicon.svg`	Scalable Vector Graphic	Hospital caduceus cross icon rendered in browser tabs.
`public/icons.svg`	SVG Sprite Sheet	UI vector iconography for medical tools, stethoscopes, pills, and checkmarks.
`src/assets/hero.png`	High-Res Raster Image	Hero visual artwork showcasing mobile and desktop clinic interfaces.

FILE PATH	TYPE / FORMAT	ROLE IN APPLICATION
`src/assets/react.svg` & `vite.svg`	SVG Vector Logos	Vite and React framework branding assets.
`ArogyaNext_Pitch_Deck.html`	Presentation Deck HTML	Executive slide deck outlining market opportunity, traction, and ROI for investors.

System Integrity Verification: All files adhere to strict modular separation. State updates propagate reactively via Firestore `onSnapshot` listeners, delivering real-time, sub-second synchronization across all 7 operational interfaces.